16 Prove algebraically that $0.6\dot{3}$ can be written as $\frac{19}{30}$

Let $t = 0.633333...$

$10t = 6.333333...$

1 mark
Align the repeating tails by multiplying by 10.

Subtract: $10t - t = 6.333333... - 0.633333...$

$9t = 5.7$, so $90t = 57$

$t = \frac{57}{90} = \frac{19}{30}$

1 mark
The repeating tails cancel exactly.

(Total for Question 16 is 2 marks)

DO NOT WRITE IN THIS AREA